전해정련을 통한 선택적 산화된 폐 영구자석 분말 내 금속 회수

Metal Recovery in Selectively Oxidated Permanent Magnet Waste Powder by Electrorefining

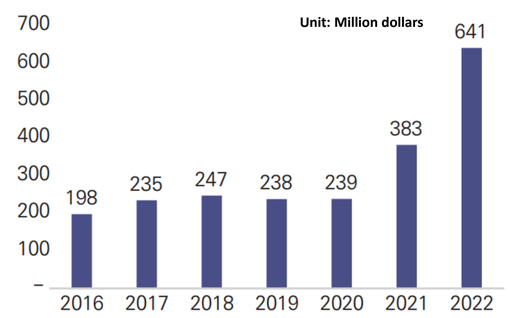
허현재, 윤재우, 김선정*

울산대학교 신소재·반도체융합학부

*sunjungkim@ulsan.ac.kr

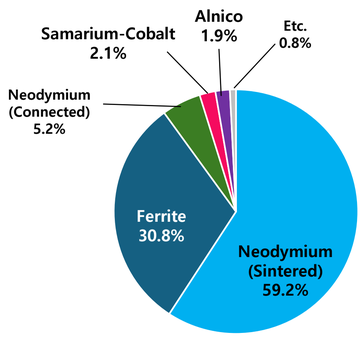
Introduction

❖ **Tendency to import permanent magnets**



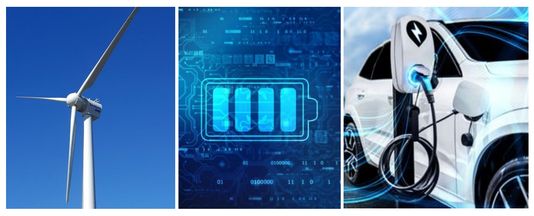
✓ 자료 : Uncomtrade, 한국무역협회

❖ **Market share by permanent magnet**



✓ 자료 : J UN CUI et al.(2022.4)

❖ **Application of Nd-Fe-B Permanent Magnet**

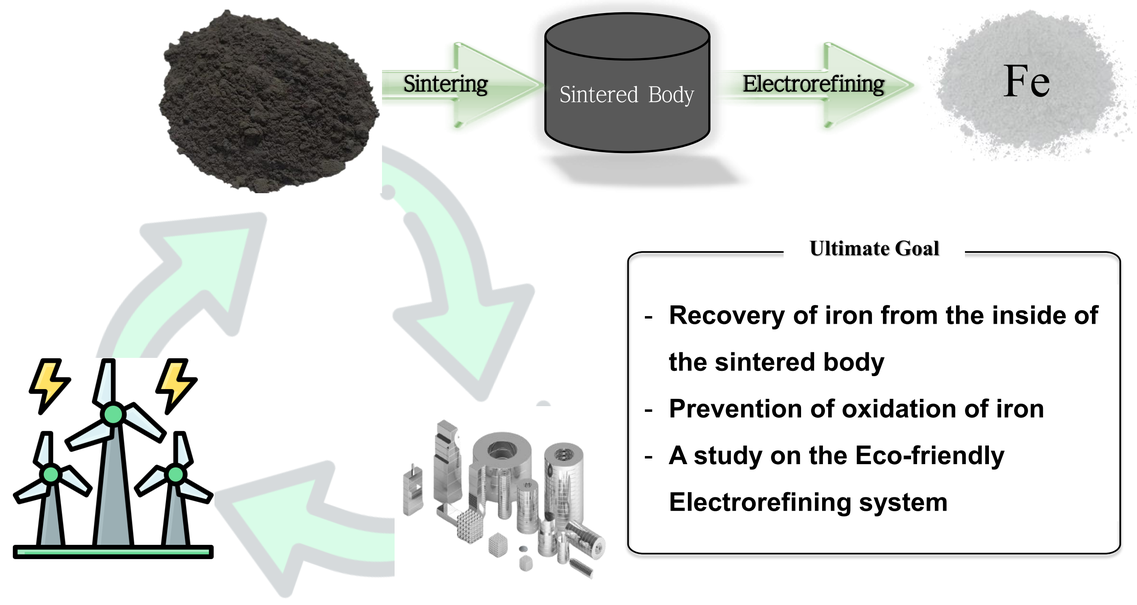


❖ **Causing environmental pollution during smelting and processing**

✓ Neodymium-based permanent magnets are essential components in clean energy technologies such as wind turbines, electric vehicles, and battery systems.
✓ Due to their high magnetic performance and growing global demand, the import of rare earth magnets has surged, highlighting their strategic and economic importance.
✓ However, conventional smelting processes generate significant environmental concerns, necessitating sustainable recovery methods.

Research Goal

❖ **Mitigate supply chain risk and recover segregation**



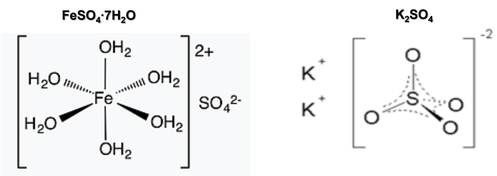
✓ The goal of this study is to build technology for recycling in an eco-friendly way to mitigate the supply chain risk of neodymium permanent magnets.
✓ After powdering the waste neodymium permanent magnet, the sintered body was manufactured through a sintering process, and the iron inside the sintered body was recovered in the form of pure unoxidized iron through an electrolytic refining process.

Experimental Method

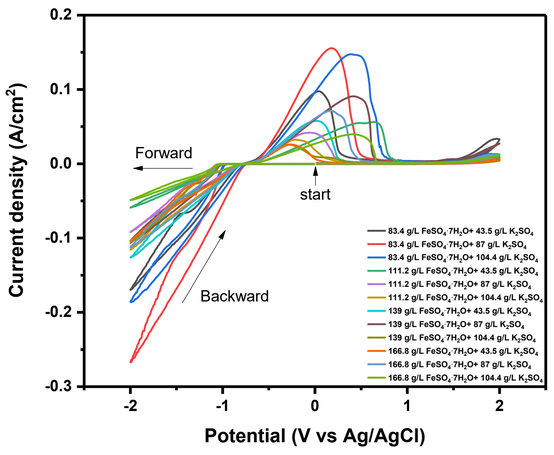
❖ **Variables**

➢ Composition of Electrolyte

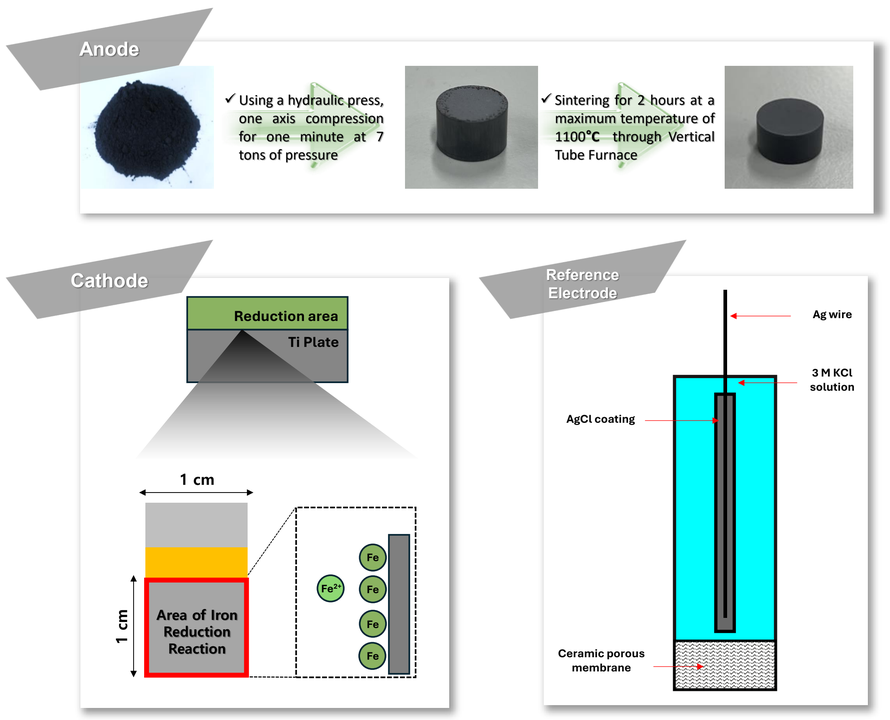
$FeSO_4 \cdot 7H_2O$ [g/L]	K_2SO_4 [g/L]	$FeSO_4 \cdot 7H_2O$ [g/L]	K_2SO_4 [g/L]
83.4	43.5	139	43.5
	87		87
	104.4		104.4
111.2	43.5	166.8	43.5
	87		87
	104.4		104.4



➢ Cathodic potential

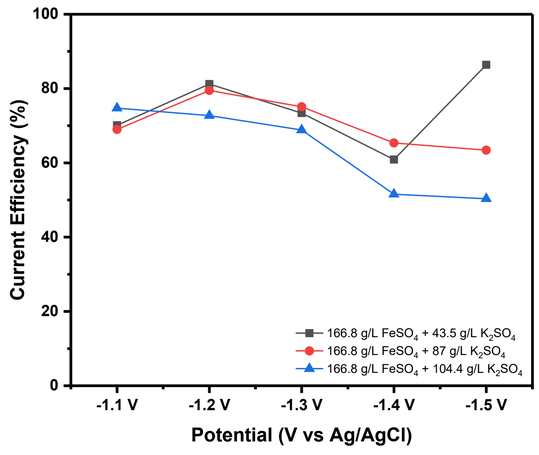


❖ **Three-electrode cell**



Results & Discussion

❖ **Current Efficiency (Faraday's Law)**



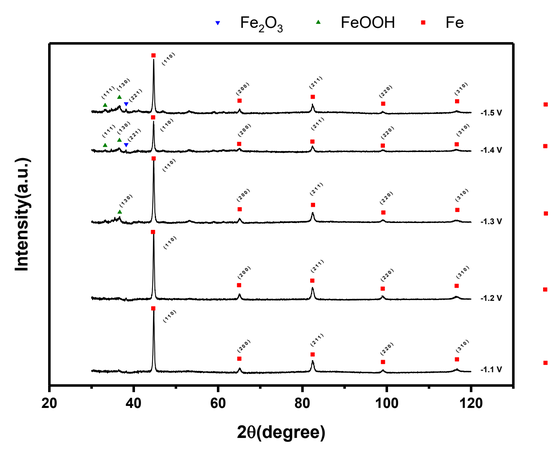
$$m = \frac{Ita}{nF}$$

m = Theoretical deposition weight [g]
I = Current [C/s]
t = Time [s]
a = Atomic weight [g/mol]
n = Number of electrons
F = Faraday's constant [96500 C/mol]

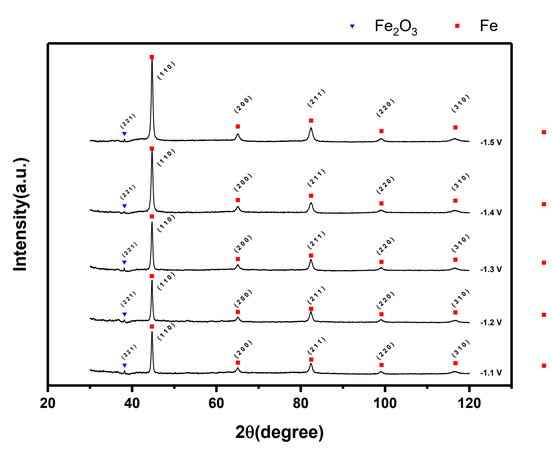
$$\text{Current Efficiency [\%]} = \frac{\text{Actual deposition weight [g]}}{\text{Theoretical deposition weight [g]}} \times 100$$

✓ Calculate the theoretical deposition wight from the atomic weight of iron.
✓ In the experiment with the highest concentration of K_2SO_4, the current efficiency was measured to be the lowest because the deposition was not well performed due to the high conductivity, and the deposition amount was measured to be low.

❖ **Phase Analysis (X-ray diffraction)**

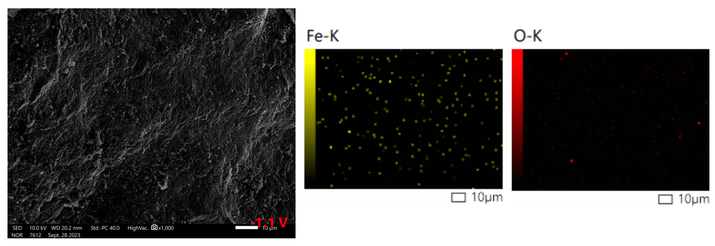


83.4 g/L $FeSO_4 \cdot 7H_2O$ + 43.5 g/L K_2SO_4

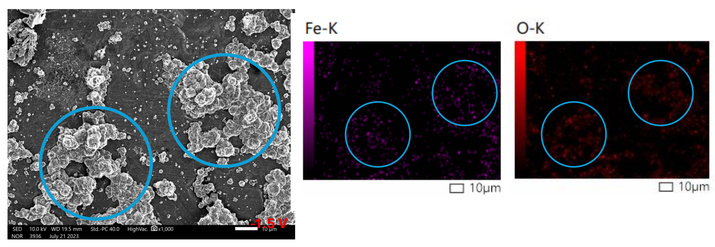


166.8 g/L $FeSO_4 \cdot 7H_2O$ + 43.5 g/L K_2SO_4

❖ **Surface Analysis (Scanning Electron Microscope)**



Element	Line	Mass %	Atom %
O	K	9.21	26.16
Fe	K	90.79	73.84



Element	Line	Mass %	Atom %
O	K	55.37	53.61
Fe	K	40.12	33.32
Ti	K	4.51	26.16

Conclusion

- The concentration of K_2SO_4, which is an electrolyte component, is proportional to the conductivity of the electrolyte.
- Experiments conducted at high potential promote the reaction of hydrogen gas to promote the oxidation of metal iron and the reduction of iron hydroxide.
- The optimal conditions were judged to be 166.8 g/L $FeSO_4 \cdot 7H_2O$ + 43.5 g/L K_2SO_4.
- Further experiments are planned by adding additives to the electrolyte to suppress the hydrogen gas reaction and control the reduction of iron hydroxide in the future.